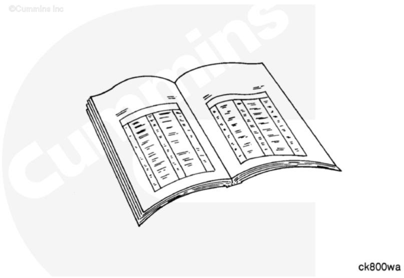
Trainings

Preparatory Steps

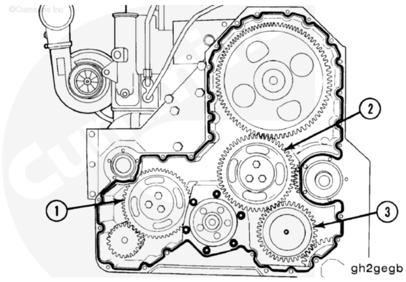
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html)



Remove

Three idler gear assemblies are used:

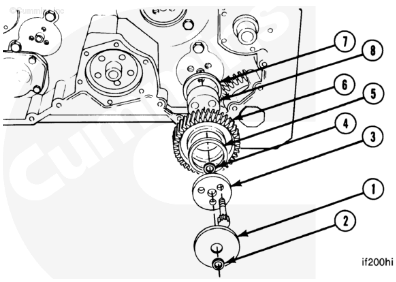
- Water pump idler gear (1)
- Camshaft idler gear (2)
- Hydraulic pump idler gear (3).



SAE 'A' and SAE 'B' Removal

To remove the hydraulic drive idler gear assembly, remove:

- Mounting spacer (1)
- Rectangular seal (2)
- The three twelve-point capscrews
- Gear retainer (3)
- Rectangular seal (4)
- Front thrust bearing (5)
- Idler gear (6)
- Rear thrust bearing (7)
- Idler gear shaft (8).

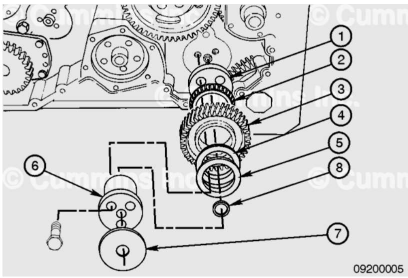


Note : SAE "A" drives do not use a rectangular seal (4) on the rear surface of the gear retainer.

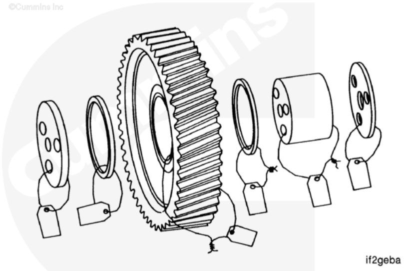
SAE 'B-B' Removal

To remove SAE B-B hydraulic drive idler gear assembly, remove:

- Mounting Spacer (1)
- Tapered Roller Bearing (2)
- Gear Drive (3)
- Tapered Roller Bearing (4)
- Shim(s) (5)
- Idler Gear Shaft (6)
- Mounting Spacer (7)
- Rectangular Seal (8).



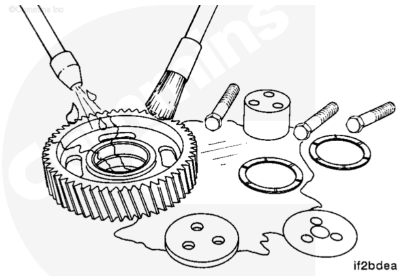
Mark or tag the individual pieces of the idler gear assembly and attach them together.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

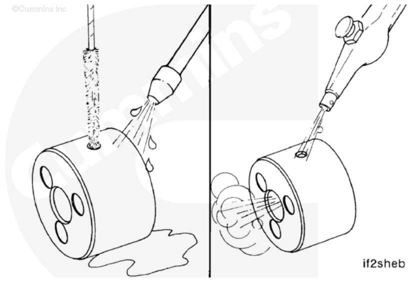
Clean the parts with solvent. Dry with compressed air.

⚠ WARNING ⚠

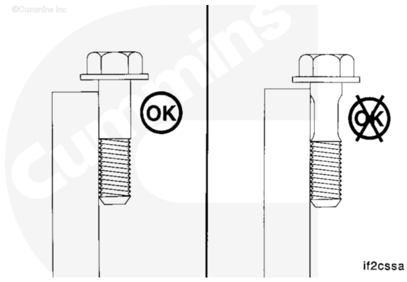
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a bristle brush to clean the oil drillings in the idler gear shaft.

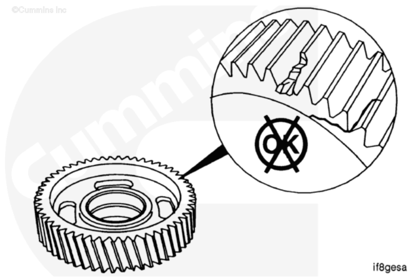
Blow out the oil drillings with compressed air.



Use a straightedge to inspect the retaining capscrews for necking. If necking is observed, do **not** reuse the capscrews. They **must** be replaced.



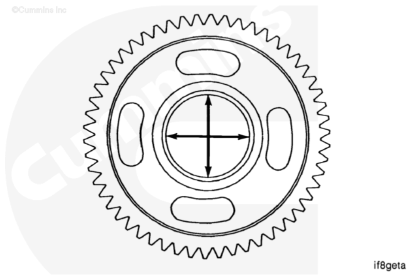
Inspect the idler gear for chipped, broken, or cracked teeth.



Note : Measurements do not apply to SAE 'B-B' hydraulic drive idler gear. Visually inspect the inside diameter of the gear for reuse.

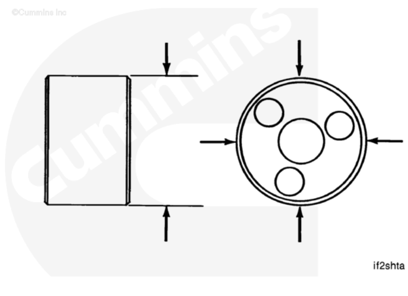
Measure the inside diameter of the idler gear bushing bore.

SAE 'A' and 'B' Hydraulic Drive Idler Gear Bushing Bore Inside Diameter		
mm		in
60.045	MIN	2.3640
60.100	MAX	2.3661



Measure the outside diameter of the idler gear shaft.

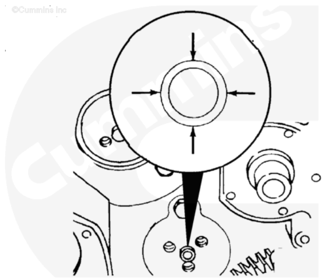
SAE 'A' and 'B' Hydraulic Drive Idler Gear shaft Outside Diameter		
mm		in
59.975	MIN	2.3612



SAE 'A' and 'B' Hydraulic Drive Idler Gear shaft Outside Diameter		
mm		in
60.008	MAX	2.3625

SAE 'B-B' Hydraulic Drive Idler Gear Shafts Outside Diameter		
mm		in
63.455	MIN	2.4982
63.486	MAX	2.4994

Measure the outside diameter of the ring dowel.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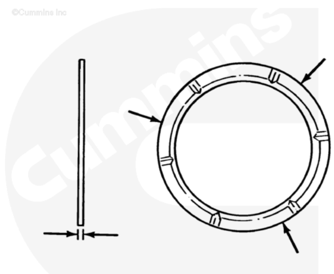
Hydraulic Drive Idler Ring Dowel Outside Diameter		
mm		in
19.217	MIN	0.7566
19.243	MAX	0.7576

Note : SAE 'B-B' hydraulic drive gear assembly does not use thrust washers.

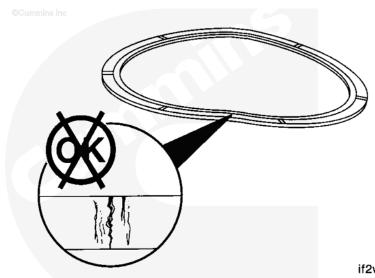
Measure the thrust washer thickness in three places 120-degrees apart.


if2wata

Hydraulic Drive Idler Thrust Washer Thickness		
mm		in
2.400	MIN	0.0945
2.470	MAX	0.0972

Note : SAE 'B-B' hydraulic drive gear assembly does not use thrust washers.

Flex the thrust washers approximately 3 to 6 mm [1/8 to 1/4 in], and inspect the surfaces for cracks.

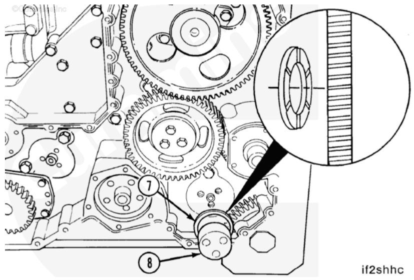


if2wasa

Install

⚠ CAUTION ⚠

The grooved side of the rear thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.



SAE 'A' and SAE 'B' Install

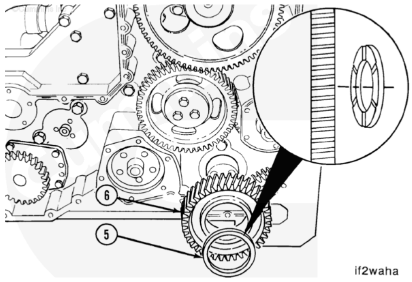
Use Lubriplate™ 105, or equivalent, to lubricate the thrust bearings and idler gear.

Install the idler gear shaft (8) and rear thrust bearing (7).

Note : When an SAE B drive is used, a special hydraulic drive idler shaft with two oil holes is used. With the engine in the upright position, orientate the shaft so that one oil hole is at twelve- o'clock and the other at four-o'clock.

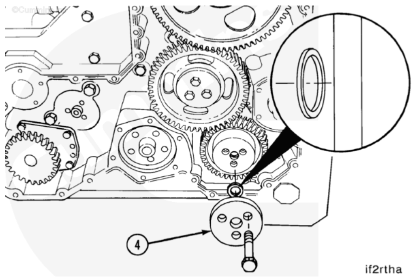
⚠ CAUTION ⚠

The grooved side of the front thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.



Install the idler gear (6) and front thrust bearing (5).

Note : On SAE "B" drives, install a new rectangular seal on the rear surface of the gear retainer (4) before installing the retainer.



Install the gear retainer (4).

Install the three retaining capscrews and tighten.

Torque Value:

1. 61 n•m [45 ft-lb]
2. Rotate 60 degrees

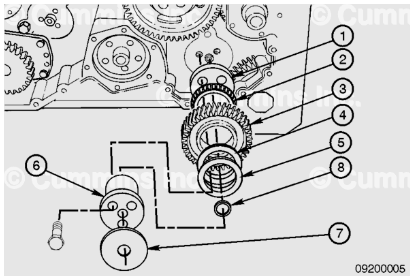
SAE 'B-B' Install

Note : Use shim(s) as necessary for proper idler gear end clearance.

Use Lubriplate™ 105, or equivalent to lubricate the roller bearings and gear.

To install SAE B-B hydraulic drive idler gear assembly, install

- Mounting Spacer (1)
- Tapered Roller Bearing (2)
- Gear Drive (3)
- Tapered Roller Bearing (4)
- Shim (s) (5)
- Idler Gear Shaft (6)
- Mounting Spacer (7)
- Rectangular Seal (8).



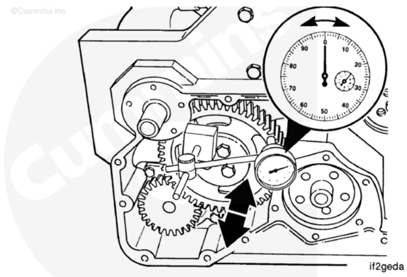
Torque Value:

1. 61 n•m [45 ft-lb]
2. Rotate 60 degrees

The following idler gear end clearance and backlash checks apply to all three idler gears. The same MINIMUM and MAXIMUM values apply, also.

Use a dial indicator gauge with a magnetic base to measure the idler gear end clearance.

Place the contact tip of the gauge against the face of the idler gear.

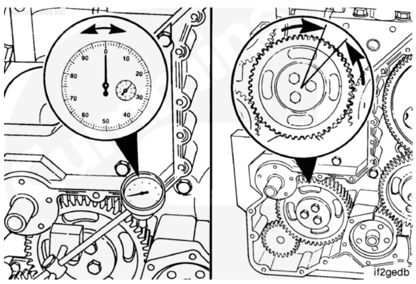


Idler Gear End Clearance		
mm		in
0.30	MIN	0.012
0.53	MAX	0.021

Use a dial indicator gauge with a magnetic base to measure the idler gear backlash.

Place the contact tip of the gauge against a tooth on the idler gear.

Do **not** allow the mating gears to move while measuring the backlash.



Idler Gear Backlash		
mm		in
0.08	MIN	0.003
0.38	MAX	0.015

Finishing Steps

- Install the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html)
- Operate the engine to normal operating temperature and check for leaks.

Last Modified: 23-Jan-2009